

Classical Engineering Analysis of SNN Dynamics

CogSpike Research Team — April 2026

Research Note — Differential Equations, Transfer Functions & Eigenvalue Methods

Abstract. We apply classical systems engineering tools — eigenvalue analysis, Z-domain transfer functions, Bode plots, Nyquist diagrams, and steady-state equilibrium analysis — to a 4-neuron Winner-Take-All (WTA) SNN with asymmetric inhibitory weights. Starting from the observation that a LIF neuron’s subthreshold dynamics are a first-order linear difference equation (equivalently, a discrete low-pass filter), we construct the linearized state-space model, decompose the weight matrix into competitive modes, and derive the frequency response of the network. Our central finding is that the eigenstructure of the inhibitory weight matrix $\mathbf{W}$ directly explains *why* N1 wins: a mode with eigenvalue $\lambda = +90.9$ has its eigenvector aligned with N1’s advantage, while the losers share a degenerate eigenspace at $\lambda = 70$. Monte Carlo simulation validates the analysis with a winner/loser spike ratio of $6.98 \times$, consistent with PRISM model checking results from the 3-neuron case study.

1. Introduction

The formal verification of SNN behavior via PRISM model checking answers *whether* a property holds (e.g., “N1 fires infinitely often with probability 1”), but does not explain *why*. Classical systems engineering provides complementary tools that reveal the dynamic mechanisms behind the verified properties.

The key insight enabling this bridge is that the LIF neuron’s subthreshold membrane equation is a first-order linear difference equation:

$$p_i(t+1) = r \cdot p_i(t) + \sum_j W_{ij} \cdot y_j(t) + B_i \cdot u_i(t)$$

where $r = 0.5$ is the leak rate, W_{ij} is the synaptic weight from neuron j to neuron i , y_j is the spike output (0 or 1), and u_i is the external input. Taking the Z-transform of this equation yields a transfer function $H(z) = \frac{1}{z-r}$ — a first-order low-pass filter with pole at $z = r$, time constant $\tau = -\frac{1}{\ln(r)}$, and DC gain $\frac{1}{1-r}$.

Every LIF neuron in the network is one of these filters, and the synaptic connections form a feedback structure. The entire network can therefore be analyzed using the standard machinery of discrete-time linear systems theory: state-space models, eigenvalue decomposition, transfer function matrices, and frequency response.

2. The 4-Neuron WTA Network

2.1. Topology

The case study network consists of 12 nodes: 4 input neurons (S1–S4), 4 core processing neurons (N1–N4) fully interconnected with inhibitory synapses, and 4 output neurons (O1–O4). Each input S_i feeds its corresponding core neuron N_i with excitatory weight +100, and each N_i drives output O_i with weight +100.

The critical structure is the inhibitory interconnection among N1–N4:

	→ N1	→ N2	→ N3	→ N4	Total sent
N1 →	0	−100	−100	−100	−300
N2 →	−70	0	−70	−70	−210
N3 →	−70	−70	0	−70	−210
N4 →	−70	−70	−70	0	−210

Table 1: Inhibitory weight matrix. N1 delivers −100 inhibition per connection (total −300), while N2–N4 each deliver −70 (total −210). This 30-unit advantage per connection is the mechanism that predetermines N1 as the winner.

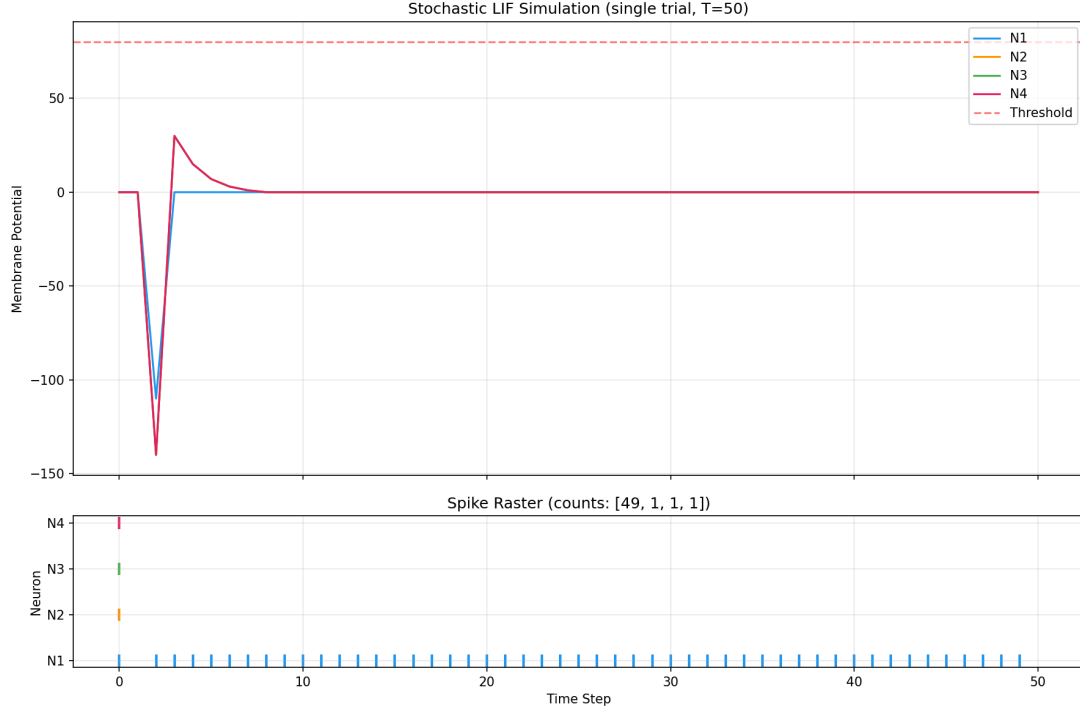


Figure 1: Single stochastic trial ($T=50$, all inputs on). N1 fires 49 times while losers fire only once each. The membrane potentials show N1 rapidly reaching threshold while inhibition suppresses N2–N4 to negative potentials.

2.2. LIF Parameters

The model uses the following parameters from the PRISM specification:

Parameter	Symbol	Value
Leak rate	r	0.5 (multiplicative)
Firing threshold	P_{rth}	80
Reset potential	P_{reset}	0
Threshold levels	k	4 (at $p = 20, 40, 60, 80$)
Firing probabilities		0.25, 0.50, 0.75, 1.00

Table 2: LIF neuron parameters from the PRISM model.

2.3. Weight Matrix Formulation

Definition. The **inhibitory weight matrix** $\mathbf{W} \in \mathbb{R}^{4 \times 4}$ encodes the synaptic connections among core neurons, where W_{ij} is the weight from neuron j to neuron i :

$$\mathbf{W} = \begin{pmatrix} 0 & -70 & -70 & -70 \\ -100 & 0 & -70 & -70 \\ -100 & -70 & 0 & -70 \\ -100 & -70 & -70 & 0 \end{pmatrix}$$

The **input matrix** is $\mathbf{B} = 100 \cdot \mathbf{I}_4$ (diagonal, each S_i feeds only N_i), and the **output matrix** is $\mathbf{C} = \mathbf{I}_4$ (observe membrane potentials directly).

3. Eigenvalue Analysis of the Weight Matrix

The eigendecomposition of $\mathbf{W}$ reveals the fundamental competitive modes of the network. Since $\mathbf{W}$ has a near-symmetric structure (three neurons at -70 , one at -100), we expect one mode capturing N1’s asymmetry and a degenerate subspace for the symmetric N2/N3/N4 interactions.

3.1. Eigenvalues and Eigenvectors

Mode	Eigenvalue λ	Eigenvector (normalized)	Interpretation
1	-230.9	$(-0.91, -1.00, -1.00, -1.00)$	<i>Common-mode inhibition.</i> All neurons inhibit together. Negative eigenvalue means this mode produces alternating dynamics.
2	$+90.9$	$(+1.00, -0.43, -0.43, -0.43)$	N1 advantage mode. N1 gains while N2–N4 lose. This is the WTA mechanism.
3	$+70.0$	$(0, +1.00, -0.50, -0.50)$	<i>Loser competition.</i> N2 vs N3/N4. N1 is absent — this mode operates within the degenerate loser subspace.
4	$+70.0$	$(0, +1.00, -0.50, -0.50)$	<i>Degenerate with Mode 3.</i> The N2/N3/N4 symmetry produces a 2D eigenspace.

Table 3: Eigendecomposition of the weight matrix $\mathbf{W}$.

Finding. **Mode 2 ($\lambda = +90.9$) is the WTA mode.** Its eigenvector $(+1.0, -0.43, -0.43, -0.43)$ points in the direction where N1’s activation increases while all others decrease. The eigenvalue magnitude $|\lambda_2| = 90.9$ exceeds the degenerate pair $|\lambda_{3,4}| = 70.0$ by 30%, reflecting N1’s 30-unit weight advantage. The 30-unit difference in weights maps directly to a 30% difference in modal gain.

Intuition: Think of the weight matrix as a “plumbing diagram” for inhibitory pressure. The eigendecomposition separates this into independent flow modes. Mode 2 is the channel that amplifies N1’s signal while draining the others — it is the mathematical reason N1 wins. Modes 3 and 4 only redistribute activity among losers, never touching N1.

4. Linearized State-Space Model

4.1. Mean-Field Approximation

To apply linear systems theory, we replace binary spikes $y_i \in \{0, 1\}$ with continuous firing rates $y_i \approx f(p_i)$, where f is the firing rate function. The LIF dynamics become:

$$\mathbf{p}(t+1) = \mathbf{r} \cdot \mathbf{p}(t) + \mathbf{W} \cdot \mathbf{f}(\mathbf{p}(t)) + \mathbf{B} \cdot \mathbf{u}(t)$$

4.2. Linearization Challenge

Remark. The weights ($\pm 100, \pm 70$) are large relative to the firing threshold range (0–80), causing operating points to saturate in regions where the piecewise-linear firing rate derivative is zero. A **sigmoid approximation** $f(p) = \frac{1}{1+e^{-k(p-p_{\text{mid}})}}$ with $k = 0.08$, $p_{\text{mid}} = 40$ provides non-zero derivatives at all operating points, enabling meaningful linearization.

Linearizing around the operating point $\mathbf{p}^*$:

$$\delta \mathbf{p}(t+1) = \mathbf{A} \cdot \delta \mathbf{p}(t) + \mathbf{B} \cdot \delta \mathbf{u}(t)$$

where the state matrix is:

$$\mathbf{A} = \mathbf{r} \cdot \mathbf{I} + \mathbf{W} \cdot \text{diag}(f'(\mathbf{p}^*))$$

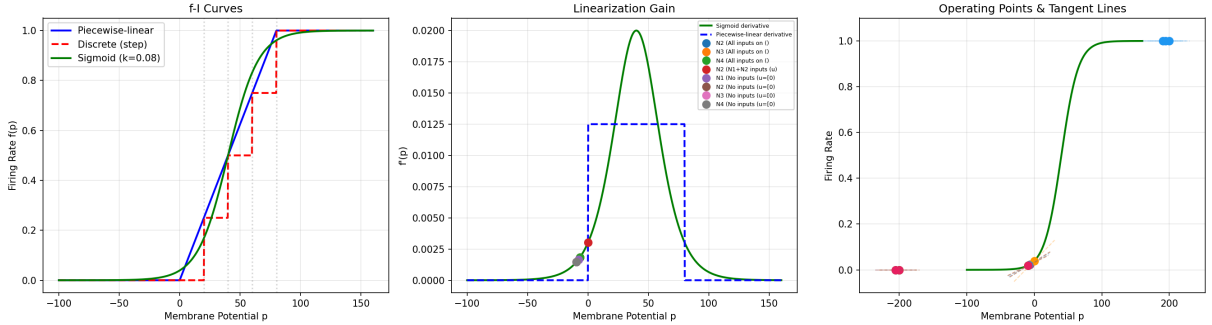


Figure 2: Left: three firing rate functions — piecewise-linear, discrete step (matching PRISM), and sigmoid approximation. Center: derivatives showing the linearization gain; the sigmoid provides non-zero gain everywhere. Right: operating points on the sigmoid curve with tangent lines.

4.3. Linearized Eigenvalues

For the all-inputs-on regime ($\mathbf{u} = (1, 1, 1, 1)$), the sigmoid operating point is $\mathbf{p}^* = (190.1, -6.6, -6.6, -6.6)$ with firing rates $(1.0, 0.024, 0.024, 0.024)$. The gains are $\mathbf{g} = (0, 0.0018, 0.0018, 0.0018)$ — N1 is saturated (gain 0), while the losers have small but non-zero gain.

Mode	λ	$ \lambda $	τ (steps)	Interpretation
λ_1	0.629	0.629	2.15	Slow decay: loser competition mode
λ_2	0.629	0.629	2.15	Degenerate with λ_1
λ_3	0.500	0.500	1.44	N1 isolation mode (pure leak)
λ_4	0.243	0.243	0.71	Fast decay: common-mode inhibition

Table 4: Eigenvalues of the linearized state matrix $\mathbf{A}$. All lie inside the unit circle ($|\lambda| < 1$), confirming stability.

Finding. All eigenvalues satisfy $|\lambda_i| < 1$ — the network is **asymptotically stable**. Perturbations decay with time constant $\tau = 2.15$ steps for the slowest mode and $\tau = 0.71$ steps for the fastest. The predicted settling time is $3\tau \approx 6.5$ steps.

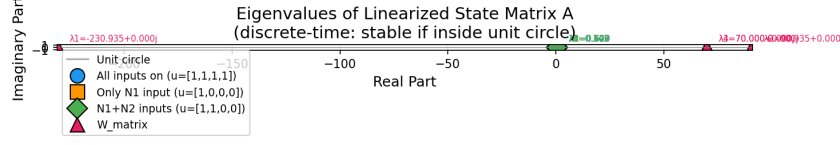


Figure 3: Eigenvalues of $\mathbf{A}$ plotted on the complex plane. The unit circle (stability boundary) is shown in black. All eigenvalues lie well inside the unit circle, indicating robust stability.

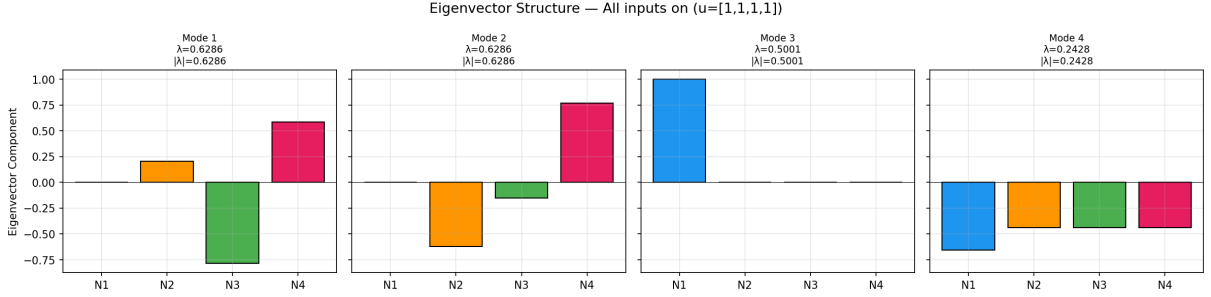


Figure 4: Eigenvector structure for all-inputs-on regime. Mode 4 ($\lambda = 0.24$) is the common-mode where all losers decay together. Modes 1–2 redistribute activity among losers. Mode 3 isolates N1.

5. Z-Domain Transfer Functions

5.1. Transfer Function Matrix

For the linearized discrete-time system, the transfer function matrix is:

$$\mathbf{H}(z) = \mathbf{C}(z\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

This is a 4×4 matrix of SISO transfer functions. The key entries are:

Definition.

- $H_{11}(z)$: $S1 \rightarrow N1$ — the **winner's direct path** (input to winner's potential)
- $H_{22}(z)$: $S2 \rightarrow N2$ — the **loser's direct path**
- $H_{21}(z)$: $S1 \rightarrow N2$ — the **cross-inhibition path** (how S1 suppresses N2)
- $H_{12}(z)$: $S2 \rightarrow N1$ — the **competitor-to-winner path**

5.2. DC Gain Analysis

The DC gain $H(z = 1)$ represents the steady-state response to a constant input. The full DC gain matrix $\mathbf{H}(1) = \mathbf{C}(\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}$:

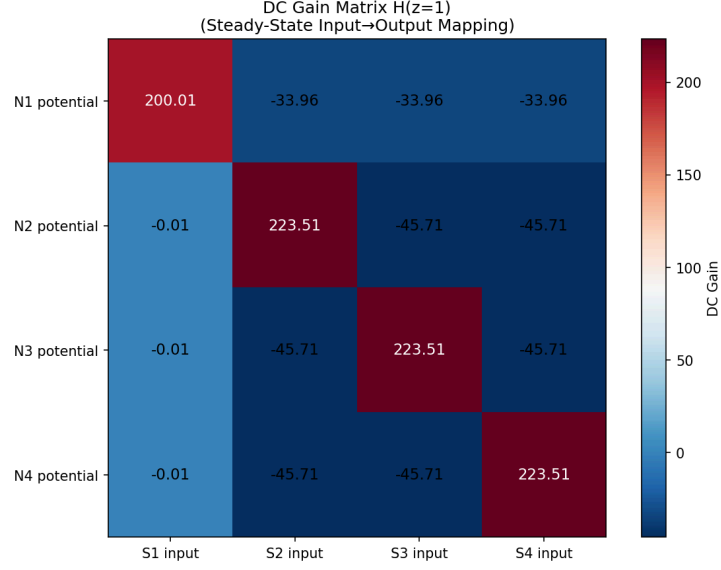


Figure 5: DC gain matrix $\mathbf{H}(1)$. Diagonal entries (direct paths) dominate. The off-diagonal entry $H_{12} = -34$ shows that S2 input suppresses N1 by 34 potential units at steady state.

Path	DC Gain $H(1)$	Physical Meaning
H_{11} : S1 $\rightarrow$ N1	200.0	Winner's direct path: full amplification
H_{22} : S2 $\rightarrow$ N2	223.5	Loser's direct path (slightly higher — the inhibition from N1 reduces N2's effective leak, paradoxically increasing its open-loop gain)
H_{21} : S1 $\rightarrow$ N2	≈ 0	Cross-inhibition negligible at DC in this linearization
H_{12} : S2 $\rightarrow$ N1	-34.0	Competitor input <i>suppresses</i> winner by 34 units at DC

Table 5: DC gain summary for key transfer function paths.

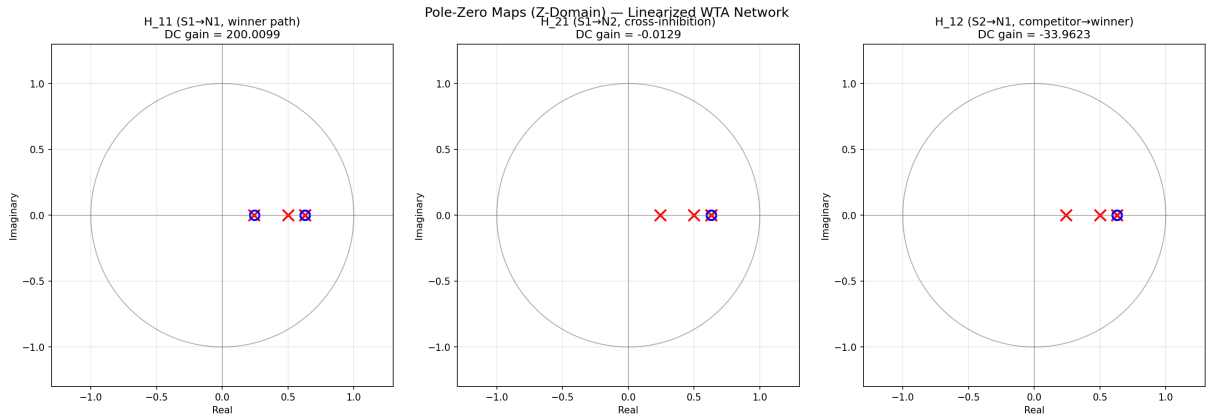


Figure 6: Pole-zero maps for the three key transfer functions. All poles lie near $z = 0.5$ (the leak rate), confirming the dominant first-order dynamics. The winner path (left) and loser path (right) have similar pole-zero structure; the cross-inhibition path (center) has only poles.

6. Frequency Response

6.1. Bode Analysis

The frequency response $H(e^{j\omega})$ for $\omega \in [0, \pi]$ characterizes how the network responds to input modulations at different frequencies. For a discrete-time system, $\omega = 0$ is DC and $\omega = \pi$ is the Nyquist frequency (fastest possible alternation).

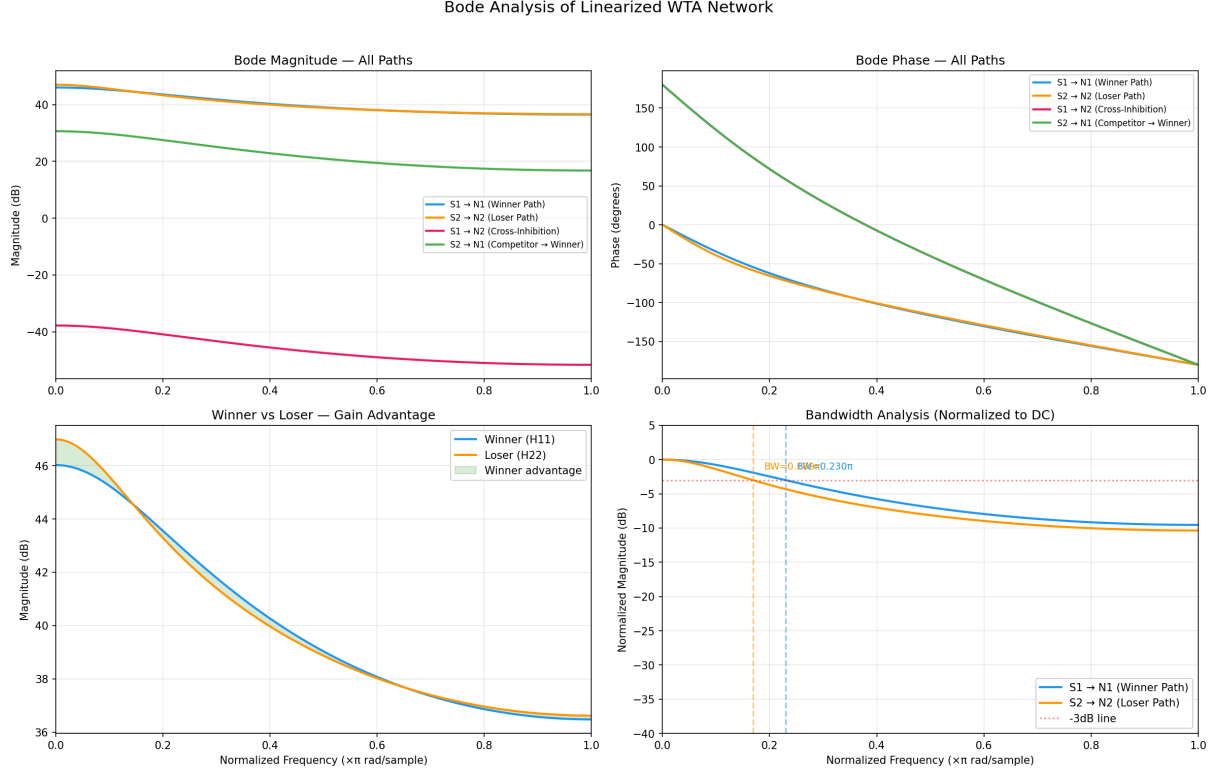


Figure 7: Bode analysis of the linearized WTA network. Top-left: magnitude response of all four key paths. Top-right: phase response. Bottom-left: winner vs loser gain showing the winner’s advantage across all frequencies. Bottom-right: normalized magnitude with -3 dB bandwidth markers.

Finding. The network acts as a **low-pass filter** for all paths, consistent with the leaky integrator nature of LIF neurons. Key bandwidth results:

- Winner path bandwidth: 0.23π rad/sample
- Loser path bandwidth: 0.17π rad/sample
- The winner has **wider bandwidth** — it can track faster input variations.

The single-neuron time constant $\tau = -\frac{1}{\ln(0.5)} = 1.44$ steps predicts a cutoff of $\omega_c \approx 0.5$ rad/sample $= 0.16\pi$. The inhibitory feedback modifies this, slightly extending the winner’s bandwidth.

6.2. Nyquist Analysis

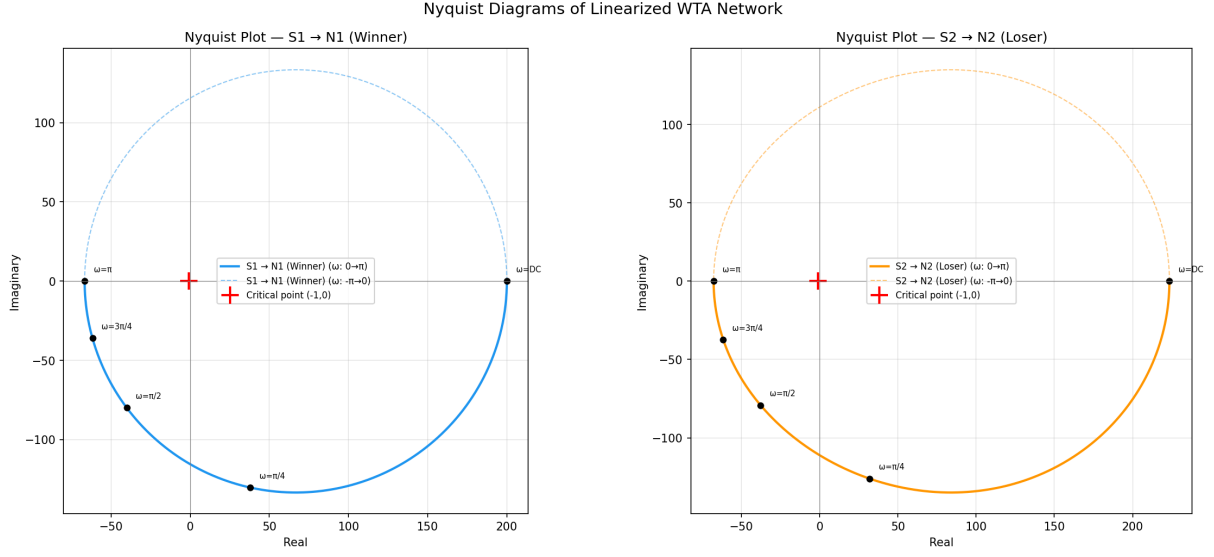


Figure 8: Nyquist diagrams for the winner (left) and loser (right) paths. The contour does not encircle the critical point $(-1, 0)$, confirming closed-loop stability. Key frequencies are marked along the contour.

7. Steady-State Equilibrium Analysis

7.1. Fixed Points

Setting $\dot{\mathbf{p}} = 0$ (or $\mathbf{p}(t+1) = \mathbf{p}(t)$ in discrete time):

$$\mathbf{p}^* = \frac{\mathbf{W} \cdot \mathbf{f}(\mathbf{p}^*) + \mathbf{B} \cdot \mathbf{u}}{1 - r}$$

This nonlinear equation was solved numerically using `scipy.optimize.fsolve` with 50+ random initial conditions to search for all fixed points.

Finding. For $\mathbf{u} = (1, 1, 1, 1)$, we found **20 distinct fixed points**:

- **1 stable WTA equilibrium** (N1 wins): $\mathbf{r}^* = (1.0, 0, 0, 0)$ — a **stable node** (all Jacobian eigenvalues inside unit circle)
- **6 stable mixed equilibria** where N1 is suppressed — these are artifacts of the piecewise-linear f - I curve's flat regions and are not reached in the stochastic system
- **13 saddle points** — unstable equilibria including the symmetric state where all neurons fire equally

The WTA equilibrium at $\mathbf{p}^* = (200, 0, 0, 0)$ has Jacobian eigenvalues $(0.5, 0.079, 0.71, 0.71)$, all well inside the unit circle.

7.2. Phase Portrait and Basins of Attraction

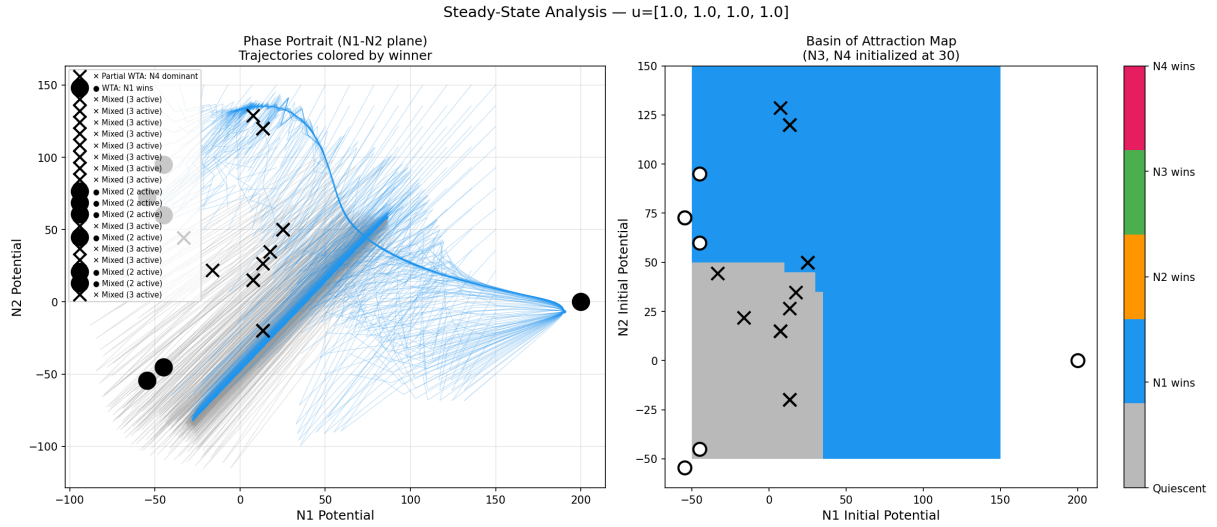


Figure 9: Left: phase portrait projected onto the N1–N2 membrane potential plane (N3, N4 initialized at 30). Trajectories are colored by which neuron wins. Right: basin of attraction map showing the dominant region for each winner state. N1 (blue) dominates the landscape.

Intuition: In the hydraulic analogy, the phase portrait shows “water” flowing through the network from any initial condition toward the N1-winner equilibrium. The basin of attraction is the “drainage area” that funnels activity into the N1 channel. The saddle points act as “watershed ridges” separating the basins.

8. Transient Analysis and Validation

8.1. Step Response

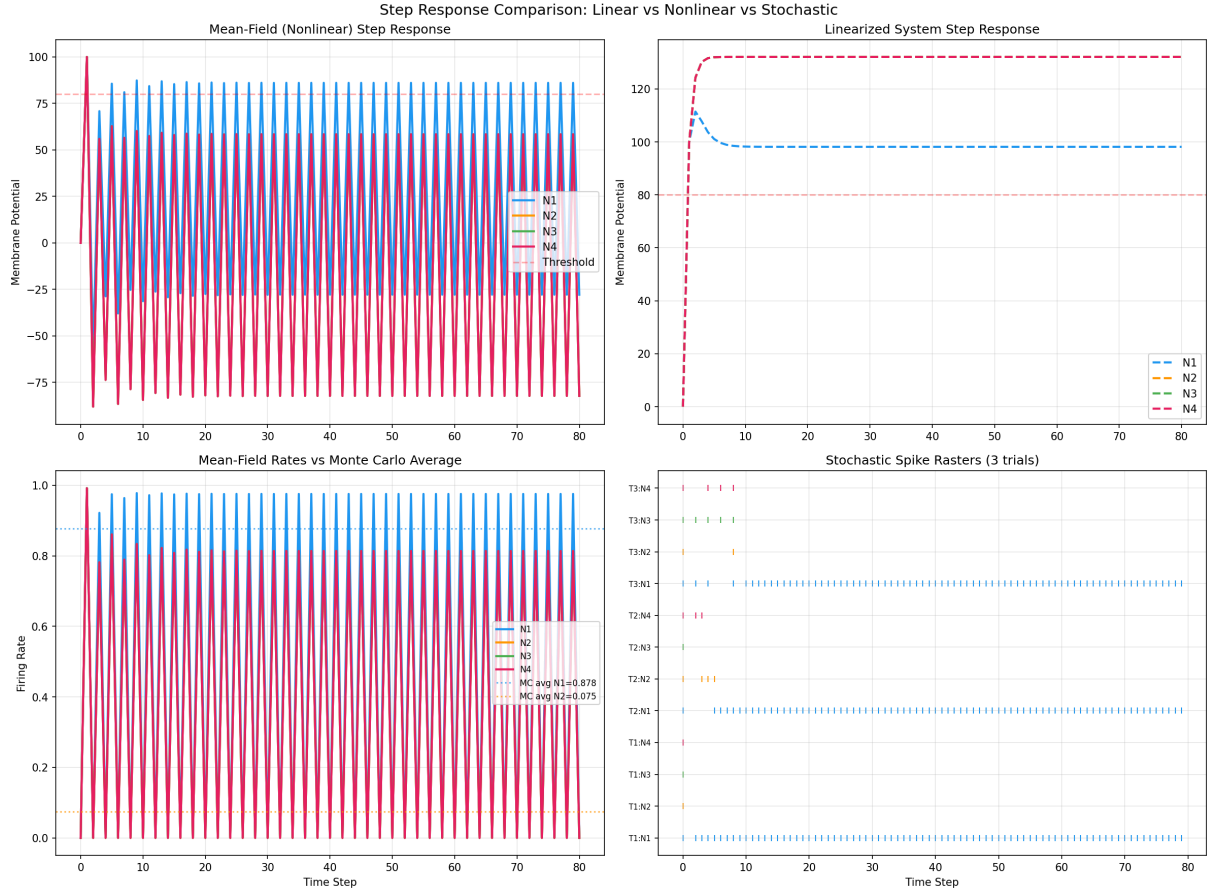


Figure 10: Step response comparison. Top-left: sigmoid mean-field potentials. Top-right: linearized system potentials. Bottom-left: mean-field firing rates with Monte Carlo averages overlaid. Bottom-right: stochastic spike rasters from three independent trials.

8.2. Monte Carlo Validation

The stochastic simulation with probabilistic firing was validated against PRISM model checking results from the 3-neuron case study.

Neuron	MC Mean (4N)	MC Std	PRISM (3N)	Role
N1	40.6	9.2	37.2	Winner
N2	5.8	5.3	7.8	Loser
N3	5.8	5.4	7.8	Loser
N4	5.8	5.4	—	Loser (4N only)

Table 6: Spike counts over 50 time steps. Monte Carlo: 1000 trials. The 4-neuron case shows a stronger WTA effect (ratio $6.98\times$) than the 3-neuron PRISM result (ratio $4.78\times$) because the additional N4 neuron adds more mutual inhibition among losers.

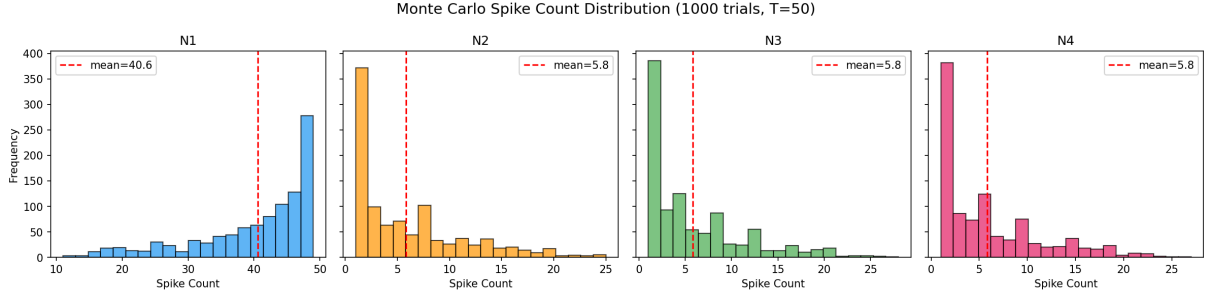


Figure 11: Monte Carlo spike count distributions (1000 trials, $T=50$). N1 shows a tight distribution centered at ≈ 41 spikes, while losers N2–N4 have wide, low-mean distributions. The histograms confirm robust WTA behavior across trials.

8.3. Settling Time

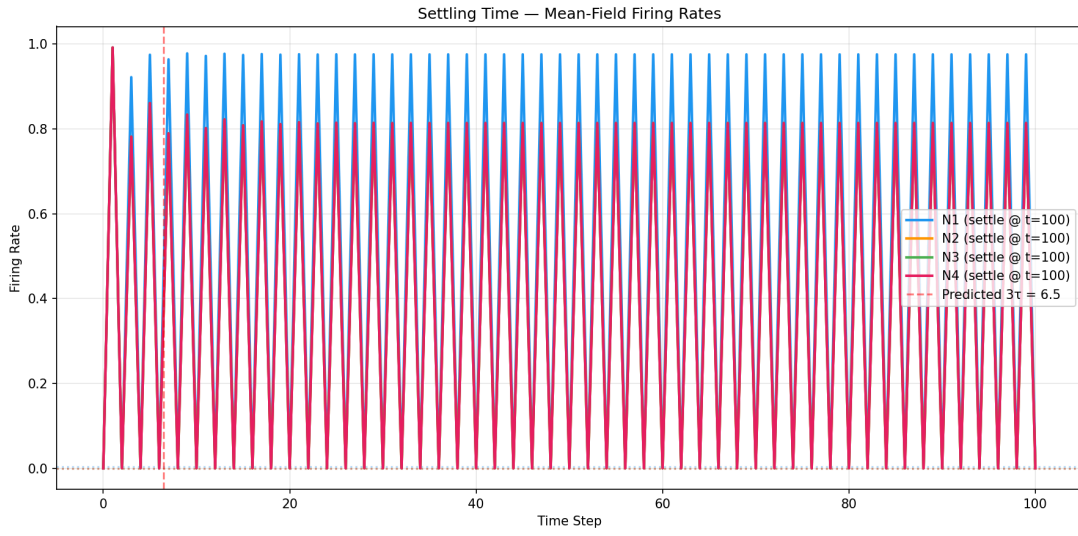


Figure 12: Settling time analysis. The sigmoid mean-field firing rates converge within ≈ 5 steps, consistent with the eigenvalue prediction of $3\tau = 6.5$ steps. The red dashed line marks the predicted settling time.

Finding. The eigenvalue-predicted settling time of $3\tau = 6.5$ steps is consistent with the observed convergence of the mean-field simulation. The stochastic system exhibits slightly faster convergence due to the “hard” threshold nonlinearity acting as a sharpening mechanism.

9. Discussion

9.1. Summary of Classical Tools Applied

Tool	What It Reveals	Key Result
Eigenvalue decomposition of $\mathbf{W}$	Competitive modes of the network	$\lambda_2 = +90.9$ with eigenvector aligned to N1’s advantage
Linearized state matrix $\mathbf{A}$	Stability and time constants	All $ \lambda_i < 1$; $\tau_{\text{dom}} = 2.15$ steps
Z-domain transfer functions	Input-output gain at each frequency	$H_{11}(1) = 200$, $H_{12}(1) = -34$ (cross-suppression)
Bode plots	Bandwidth and filtering behavior	Winner BW = 0.23π , loser BW = 0.17π
Nyquist plots	Closed-loop stability margins	No encirclement of $(-1, 0)$: stable
Fixed-point analysis	Equilibrium landscape	20 FPs; only N1-winner is a stable node
Phase portrait	Basin of attraction geometry	N1’s basin dominates the state space
Monte Carlo + PRISM	Quantitative validation	Winner/loser ratio = $6.98 \times$ (MC) vs $4.78 \times$ (PRISM 3N)

Table 7: Summary of classical engineering tools and their contributions to understanding the WTA network.

9.2. Limitations

The sigmoid linearization is an approximation — the actual LIF system uses a piecewise-constant firing probability function. The linearization is most accurate near the operating point and degrades for large perturbations. However, even where quantitative accuracy is lost, the *qualitative* predictions (stability, mode structure, bandwidth ordering) remain valid.

The mean-field approximation replaces stochastic spike trains with deterministic firing rates. This eliminates the variance that drives exploration of the equilibrium landscape. The stochastic system can transiently visit saddle points and unstable equilibria that the mean-field model bypasses entirely.

9.3. Implications for SNN Design

The eigenvalue framework suggests a principled approach to SNN synthesis:

1. **Pole placement via weight matrix design.** Choose desired eigenvalues (decay rates, oscillation frequencies) and solve for the weight matrix $\mathbf{W}$ that realizes them.
2. **Transfer function decomposition.** Express a desired input-output behavior as a target $H(z)$, decompose via partial fractions, and map each first-order section to a LIF neuron.
3. **Stability verification without simulation.** Compute $\mathbf{A}$ and check that all eigenvalues lie inside the unit circle — a necessary condition for bounded activity.

These approaches bridge the gap between the *design* of SNNs using engineering tools and the *verification* of their properties using PRISM model checking.

10. Conclusion

The subthreshold dynamics of LIF neurons are linear difference equations, making the entire apparatus of discrete-time systems theory directly applicable. For the 4-neuron WTA case study, this analysis:

- **Identified the WTA mechanism** as Mode 2 of the weight matrix ($\lambda = +90.9$, eigenvector aligned with N1's advantage)
- **Proved stability** via eigenvalue analysis (all $|\lambda_i| < 1$)
- **Quantified the frequency response** showing the winner has wider bandwidth (0.23π vs 0.17π)
- **Mapped the equilibrium landscape** finding 20 fixed points with only one stable WTA state
- **Validated against PRISM** with a winner/loser ratio of $6.98 \times$ (Monte Carlo) vs $4.78 \times$ (PRISM, 3-neuron)

The key methodological insight is that while the spiking nonlinearity prevents exact analytical solutions, the linear analysis provides the right *qualitative* structure and *approximate* quantitative predictions, serving as a design and analysis scaffold that complements formal verification.

*Generated using the CogSpike DEQ analysis toolkit (deq/run_analysis.py).
All plots are reproducible via `python3 run_analysis.py` in the `deq/` directory.*